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Portofolio - Intensive Bootcamp

Intro to Data Analytics

Owner: Fadhil Ramdani

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Problem 1

Konteks:

- Di salah satu Online Travel Agent Company kalian berperan sebagai data analyst untuk product Flight
- Ditemukan sebuah anomali bahwa terjadi lonjakan penjualan pada tanggal 3 November 2023
- Lonjakan terjadi sebesar 50% dibanding hari sebelumnya dan 25% dibanding tanggal yang sama bulan lalu
- Dilain sisi, terdapat 2 promo marketing campaign yang sedang berjalan:
  - Promo Flight 50% dengan max. 200 ribu untuk pelanggan baru
  - Promo Flight 10% dengan max 500 ribu untuk pelanggan lama
- Dilain sisi, ada promo referral yang sedang berjalan dibulan ini dimana
  - Setiap user yang join / bertransaksi untuk pertama kali, akan mendapatkan hadiah IDR 50k untuk orang yang diajak dan mengajak
- User datang ke kalian untuk mencari tahu apa penyebab terjadinya kejadian diatas

Buatkan Analytical Thinking Framework untuk case diatas yang mencakup:

- *Background*
- *Objective*
- *Data*
- *Point to be checked/ Initial Hypotheses*

Answer Form

Background

- Anomali lonjakan penjualan pada tanggal 3 November 2023
- Lonjakan sebesar 50% dibanding hari sebelumnya dan 25% dibanding tanggal yang sama bulan lalu
- Terdapat campaign dan promo yang sedang berjalan

Data

- Data penjualan dari tahun ini sampai tahun lalu
- Data pengguna campaign dan promo
- Data campaign dan promo orang lama dan baru
- Data sales sesuai tujuan

Objective

- Mengetahui kenapa terjadi lonjakan penjualan pada tanggal 3 November 2023
- Anomali penjualan disebabkan oleh promo atau campaign

Point to be checked/ Initial Hypotheses

- Anomali penjualan terjadi pada tanggal 3 November 2023 saja atau juga terjadi pada bulan lalu dengan tanggal yang sama atau tahun lalu dengan tanggal dan bulan yang sama?
- Anomali terjadi karena banyaknya promo dan campaign?
- Apakah tanggal tersebut bertepatan dengan hari libur? Dan bandingkan dengan tanggal saat tidak libur?, dan anomali terjadi juga pada hari libur yang lain?

Problem 2

Konteks:

- Di salah satu Money Transfer company
- Ditemukan rata-rata waktu tunggu transaksi mengalami lonjakan
- Hal ini menyebabkan banyak user yang complaint
- Kejadian terjadi selama periode 1-5 Nov 2023
- Belum diketahui penyebab dari insiden diatas

Buatkan Analytical Thinking Framework untuk case diatas yang mencakup:

- *Background*
- *Objective*
- *Data*
- *Point to be checked/ Initial Hypotheses*

Answer Form

Background
Terjadi lonjakan waktu tunggu selama periode 1-5 November 2023

Objective
Mengetahui penyebab terjadinya lonjakan waktu tunggu pada periode 1-5 November 2023 dan berikan rekomendasi agar waktu tunggu berkurang

Data
- Data transaksi transfer mencakup
  - nama
  - tanggal transaksi
  - dari bank apa
  - bank tujuan
  - data frekuensi transaksi

Point to be checked/ Initial Hypotheses
- Lonjakan terjadi setiap tanggal 1-5 November saja atau tanggal 1-5 setiap bulan
- Apakah server down karena transaksi banyak (dilihat dari frekuensi transaksi)
- Apakah tanggal 1-5 merupakan hari spesial(pencairan gaji atau event lain)

Problem 3

Konteks:

- Anda merupakan salah satu Marketing Data Analyst di salah satu Tech Company.
- Ditemukan bahwa terjadi penurunan jumlah akuisisi customer*, padahal secara budget iklan yang dikeluarkan cenderung stagnan
- Hal ini menyebabkan biaya akuisisi pelanggan (CAC, customer acquisition cost) menjadi lebih mahal.
- Kejadian terjadi di sepanjang pekan ke-3 dan ke-4 akhir tahun 2023 kemarin.

* akuisisi customer: jumlah customer yang mendaftar ke dalam app / platform

Buatkan Analytical Thinking Framework untuk case diatas yang mencakup:

- *Background*
- *Objective*
- *Data*
- *Point to be checked/ Initial Hypotheses*

Answer Form

Background
Pekan ke-3 dan ke-4 akhir tahun 2023 terjadi penurunan jumlah akuisisi customer dengan budget iklan stagnan menyebabkan biaya akuisisi pelanggan lebih mahal

Objective
Mengetahui penyebab penurunan jumlah akuisisi customer pada pekan ke-3 dan ke-4 akhir tahun 2023

Data
- Data waktu pekan ke-3 dan ke-4 akhir tahun 2023 dan 2 tahun sebelumnya
- Data akuisisi customer
- Data biaya akuisisi pelanggan
- Data budget iklan
- Data penyaluran budget iklan
- Data CAC

Point to be checked/ Initial Hypotheses
- Penurun juga terjadi di pekan 3 dan 4 pada tahun tahun sebelumnya
- penurun terjadi akibat akuisisi customer banyak yang tidak berhasil
- Budget iklan yang mahal disebabkan oleh penurunan akuisisi customer saja atau ada penyebab lain

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Portofolio - Intensive Bootcamp

Intro to Statistics

Owner: Fadhil Ramdani

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Standard Dev - Task

Fadhil Ramdani_MiniTask_Intro to Statistics .XLSX

Date	Visitor of the Public Service
2022-12-01	4
2022-12-02	2
2022-12-03	5
2022-12-04	10
2022-12-05	7
2022-12-06	8
2022-12-07	10
2022-12-08	5
2022-12-09	8
2022-12-10	4
2022-12-11	9
2022-12-12	6
2022-12-13	10
2022-12-14	8
2022-12-15	4
2022-12-16	6
2022-12-17	8
2022-12-18	5
2022-12-19	7
2022-12-20	3
2022-12-21	9

Case

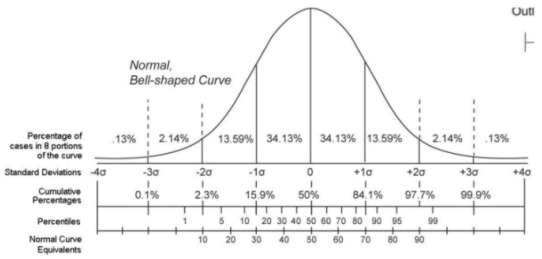
If you are the leader of one institution in Surabaya, how many chair that you need to prepared to cover 68% of all visitior at least will get the seat?

Mean	6.43
Stdev	2.14

banyak kursi yang diperlukan untuk mencover 68% = 18 kursi

mean +1*(stdev)

8.58 18.3950894



Copy of Standard Dev | Standard Dev | Z Score | Percentile | STATS | Standard Dev - Task | Z Score Task | Percentil

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Z Score Task

https://docs.google.com/spreadsheets/d/1DLEwk67jAulte0q_TM5QWZENyWw0AF0T/edit?gid=176203817#gid=176203817

Fadhil Ramdani_MiniTask_Intro to Statistics .XLSX

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Share

Q17

Summarize this data

	A	B	C
2		Biaya Hidup Bulanan	
3	Data Teman Udin	Bekasi	Tuban
4	1	4,800,000	2,500,000
5	2	5,000,000	2,100,000
6	3	4,200,000	3,000,000
7	4	4,600,000	3,000,000
8	5	4,500,000	2,500,000
9	6	5,200,000	2,400,000
11		Bekasi	Tuban
12	Tawaran Kerja dengan Gaji Kepada si Udin	5,000,000	3,500,000
14	**Mana yang harus Udin Ambil jika mempertimbangkan biaya hidup berdasar lingkungannya?**		
15	Mean Sample	4,716,667	2,583,333
16	Std. Sample	360092.5807	354494.9459
17	Z Score	0.7868346879	2.585838465
18	Z Score persen	78,23%	99,51%

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

Copy of Standard Dev | Standard Dev | Z Score | Percentile | STATS | Standard Dev - Task | Z Score Task | Percentil

Percentile - Task

Fadhil Ramdani_MiniTask_Intro to Statistics .XLSX

I6: Kondisi P90 = 10 menit adalah warning sehingga prioritas utama adalah menurunkan durasi transaksi pada desil terakhir tersebut melalui optimasi infrastruktur dan automasi proses agar sesuai

Transaction ID	service duration
1	1
2	4
3	8
4	5
5	5
6	1
7	10
8	5
9	6
10	1
11	1
12	10
13	2
14	1
15	4
16	6
17	8
18	4
19	9
20	5
21	10
22	7

Task

In one money transfer company, the expected of Percentile 90 SLA is under 5 mins to ensure the customer satisfaction of the service provided

There are 50 transactions occured with each of SLA is attached

Is the company was achieved P90 Satisfaction Level Condition? 10

What is your recommendation to product team to solve this condition?

Kondisi P90 = 10 menit adalah warning sehingga prioritas utama adalah menurunkan durasi transaksi pada desil terakhir tersebut melalui optimasi infrastruktur dan automasi proses agar sesual dengan janji layanan

P90 10

Client tidak akan membeli karna aktualnya itu 10 menit , yang dijanjikan p90 = 5 menit (atau artinya yang dijanjikan 90% transaksi yang disediakan pasti berada dibawah atau

Standard Dev | Z Score | Percentile | STATS | Standard Dev - Task | Z Score Task | Percentile - Task | STATS - Task

Percentile - Task

Fadhil Ramdani_MiniTask_Intro to Statistics .XLSX

I6: Kondisi P90 = 10 menit adalah warning sehingga prioritas utama adalah menurunkan durasi transaksi pada desil terakhir tersebut melalui optimasi infrastruktur dan automasi proses agar sesuai

Transaction ID	service duration
1	1
2	4
3	8
4	5
5	5
6	1
7	10
8	5
9	6
10	1
11	1
12	10
13	2
14	1
15	4
16	6
17	8
18	4
19	9
20	5
21	10
22	7

Task

In one money transfer company, the expected of Percentile 90 SLA is under 5 mins to ensure the customer satisfaction of the service provided

There are 50 transactions occured with each of SLA is attached

Is the company was achieved P90 Satisfaction Level Condition? 10

What is your recommendation to product team to solve this condition?

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P90 10

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Standard Dev | Z Score | Percentile | STATS | Standard Dev - Task | Z Score Task | Percentile - Task | STATS - Task

STATS - Task

Fadhil Ramdani_MiniTask_Intro to Statistics .XLSX

	A	B	C	D	E	F	G
1	uid	Voucher Redeem		Case			
2	1	98		You are a part of CRM Team and asked to evaluate the redeemed voucher of the day			
3	2	112					
4	3	1		1. What are the			
5	4	41			mean		
6	5	106			median		
7	6	111			mode		
8	7	86			std.dev		
9	8	142			and upper & lower threshold	*cari tau apakah ada outlier*	
10	9	143		of the last month performance?			
11	10	88		2. Is there any outlier from the last month performance?		500	
12	11	32		3. If yes? how many?		1	
13	12	66					
14	13	92		Answer			
15	14	94		Mean	97.9		
16	15	118		Median	89		
17	16	53		Mode	86		
18	17	119		Var	7054.92069		
19	18	500		Std Dev S	83.99357529		
20	19	37		Percentile 25	54.5		
21	20	90		Percentile 75	114.25		
22	21	128		IQR	59.75		
23	22	79					
24	23	86		LOWER THRES	-35.125		
25	24	45		UPPER THRES	203.875		
26	25	33		COUNTIF	1		
27	26	54					
28	27	115					
29	28	129					
30	29	56					
31	30	83					

Standard Dev | Z Score | Percentile | STATS | Standard Dev - Task | Z Score Task | Percentile - Task | STATS - Task

Link Google Sheets :
https://docs.google.com/spreadsheets/d/1DLEwk67jAulte0q_TM5QWZENyWw0AF0T/edit?usp=sharing&ouid=110204954342699880842&rtpof=true&sd=true

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Portofolio - Intensive Bootcamp

Introduction to Problem Solving

Owner: Fadhil Ramdani

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Mini Task

- Halojek is a company with 2 main product:
  - Halofood : a service to provide food delivery
  - Haloride : a service to provide mobility using motor cycles
- The revenue on this quarter (Q3, 23) is decreased for 50% compared to same quarter last year (Q3, 22)

Mini Task (Cont)

The cost for this quarter is as follow:

- Marketing cost : IDR 800 Bio,
- Labor cost: IDR 750 Bio
- Infrastructure cost: IDR 3,500 Bio

The cost for last quarter is as follow:

- Marketing cost : IDR 600 Bio,
- Labor cost: IDR 600 Bio
- Infrastructure cost: IDR 2000 Bio

Objective: Develop problem solving framework for the case above, you are a data analyst to deep dive the root cause of cost increasing on this quarter and also provide the alternative

Step 1: Clarify the Problem

Clue: Quantified the gap

Ideal Condition:
IDR 3,200 (Q3 2022)

Gap:
IDR 1,850 naik
57.8% dari tahun lalu
di Quarter yang sama

Existing Condition:
IDR 5,050 (Q3 2023),
sedangkan pendapatan
turun 59%

Step 2: Breakdown the Problem

Clue: Should be data driven | 4W - What, Where, Who, When

Ideal Condition:
IDR 3,200 (Q3 2022)

Existing Condition:
IDR 5,050 (Q3 2023), sedangkan pendapatan turun 59%

Gap:
IDR 1,850 naik 57.8% dari tahun lalu di Quarter yang sama

Infrastructure cost
IDR 1,500

Marketing cost:
IDR 200

Labor cost:
IDR 150

Step 3: Set the Objective

Clue: objective should be quantified and time-bound

Ideal Condition:
IDR 3,200 (Q3 2022)

Existing Condition:
IDR 5,050 (Q3 2023), sedangkan pendapatan turun 59%

Gap:
IDR 1,850 naik 57.8% dari tahun lalu di Quarter yang sama

Infrastructure cost
IDR 1,500

Marketing cost:
IDR 200

Labor cost:
IDR 150

Objective

Menurunkan biaya Infrastructure cost sebesar IDR 1,500 dengan rentang waktu 6 bulan atau Q1 2024

Step 4: Define the root cause

Clue:

- Why, why, why
- Brainstorm is allowed

Ideal Condition:
IDR 3,200 (Q3 2022)

Existing Condition:
IDR 5,050 (Q3 2023), sedangkan pendapatan turun 59%

Gap:
IDR 1,850 naik 57.8% dari tahun lalu di Quarter yang sama

Infrastructure cost
IDR 1,500

Marketing cost:
IDR 200

Labor cost:
IDR 150

Terdapatnya rekrutmen karyawan yang terlalu banyak

Terjadi kendala pada pengiriman logistic sehingga harus dilakukan perawatan yang berakhir dengan pembekaan biaya

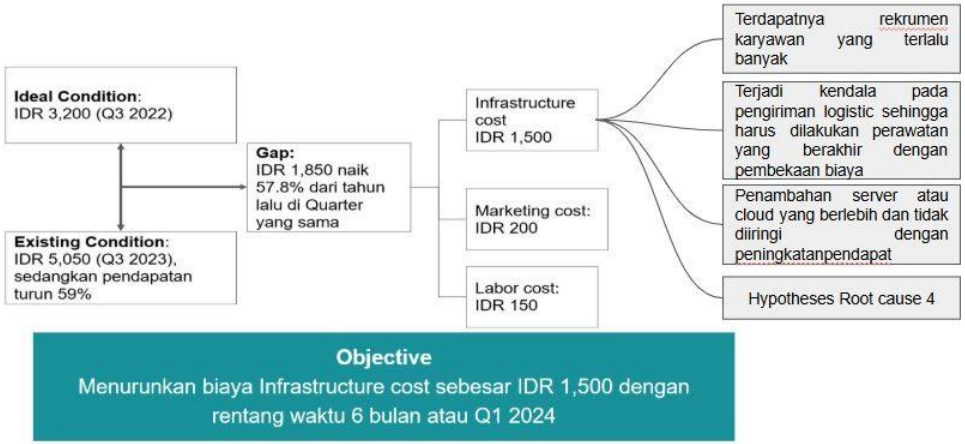
Penambahan server atau cloud yang berlebih dan tidak diiringi dengan peningkatan pendapat

Hypotheses Root cause 4

Objective

Menurunkan biaya Infrastructure cost sebesar IDR 1,500 dengan rentang waktu 6 bulan atau Q1 2024

5. Develop Countermeasures



Problem (Root Cause)	Alt. Countermeasure	Complexity	Cost	Speed	Conclusion
Penambahan server/cloud berlebih +1	Melakukan audit penggunaan server dan mematikan layanan yang tidak terpakai.	☑	☑	☑	Pilih
	Migrasi ke penyedia cloud lain dengan skema harga yang lebih efisien.	☐	☐	☐	Abaikan
Biaya logistik & perawatan membengkak	Meninjau ulang kontrak dengan vendor perawatan pihak ketiga.	☑	☑	☑	Pilih
	Mengganti seluruh armada motor dengan unit baru untuk mengurangi biaya servis.	☐	☐	☐	Abaikan
Rekrutmen karyawan terlalu banyak	Melakukan hiring freeze dan optimalisasi beban kerja tim yang ada.	☑	☑	☑	Pilih
	Melakukan restrukturisasi atau pengurangan karyawan secara massal.	☐	☐	☐	Pertimbangan Akhir

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Portofolio - Intensive Bootcamp

Data Formatting and Validation

Owner: Fadhil Ramdani

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Data Formatting and Validation

1. Gunakan file ini lalu *make a copy*
2. Ubah format data pada kolom diskon menjadi percentage
3. Ubah format data pada kolom sales dan profit menjadi currency dengan mata uang dollar dan maksimal dua angka di belakang koma
4. Pada kolom profit tandai profit yang negatif dengan warna merah
5. Ubah data pada kolom Ship_Mode sampai Region dengan Proper Case
6. Periksa apakah data superstore memiliki transaksi duplikat
7. Pada bagian Segment terapkan data validation sehingga data yang masuk adalah satu dari pilihan berikut: Corporate, Consumer, Home Office

Ubah format data pada kolom diskon menjadi percentage

Jawaban yang di kotak hijau

https://docs.google.com/spreadsheets/d/1twm745LLy_OEMd1EN2oqOhmlLbAFrf15laZIvkvjldw/edit?gid=0#gid=0

minitask pertemuan 5

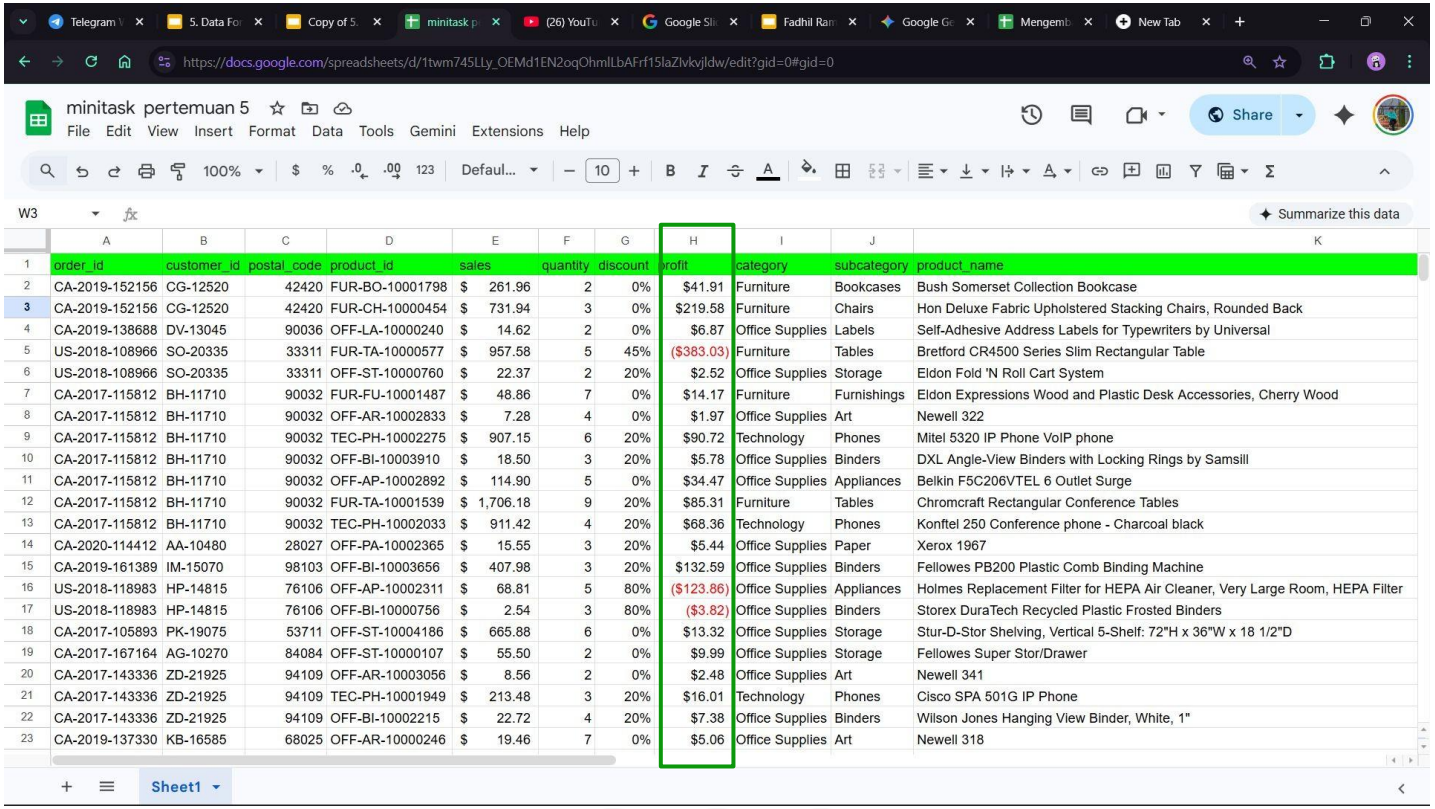
File Edit View Insert Format Data Tools Gemini Extensions Help

	A	B	C	D	E	F	G	H	I	J	K
1	order_id	customer_id	postal_code	product_id	sales	quantity	discount	profit	category	subcategory	product_name
2	CA-2019-152156	CG-12520	42420	FUR-BO-10001798	$ 261.96	2	0%	$41.91	Furniture	Bookcases	Bush Somerset Collection Bookcase
3	CA-2019-152156	CG-12520	42420	FUR-CH-10000454	$ 731.94	3	0%	$219.58	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs, Rounded Back
4	CA-2019-138688	DV-13045	90036	OFF-LA-10000240	$ 14.62	2	0%	$6.87	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters by Universal
5	US-2018-108966	SO-20335	33311	FUR-TA-10000577	$ 957.58	5	45%	($383.03)	Furniture	Tables	Bretford CR4500 Series Slim Rectangular Table
6	US-2018-108966	SO-20335	33311	OFF-ST-10000760	$ 22.37	2	20%	$2.52	Office Supplies	Storage	Eldon Fold 'N Roll Cart System
7	CA-2017-115812	BH-11710	90032	FUR-FU-10001487	$ 48.86	7	0%	$14.17	Furniture	Furnishings	Eldon Expressions Wood and Plastic Desk Accessories, Cherry Wood
8	CA-2017-115812	BH-11710	90032	OFF-AR-10002833	$ 7.28	4	0%	$1.97	Office Supplies	Art	Newell 322
9	CA-2017-115812	BH-11710	90032	TEC-PH-10002275	$ 907.15	6	20%	$90.72	Technology	Phones	Mitel 5320 IP Phone VoIP phone
10	CA-2017-115812	BH-11710	90032	OFF-BI-10003910	$ 18.50	3	20%	$5.78	Office Supplies	Binders	DXL Angle-View Binders with Locking Rings by Samsill
11	CA-2017-115812	BH-11710	90032	OFF-AP-10002892	$ 114.90	5	0%	$34.47	Office Supplies	Appliances	Belkin F5C206VTEL 6 Outlet Surge
12	CA-2017-115812	BH-11710	90032	FUR-TA-10001539	$ 1,706.18	9	20%	$85.31	Furniture	Tables	Chromcraft Rectangular Conference Tables
13	CA-2017-115812	BH-11710	90032	TEC-PH-10002033	$ 911.42	4	20%	$68.36	Technology	Phones	Konftel 250 Conference phone - Charcoal black
14	CA-2020-114412	AA-10480	28027	OFF-PA-10002365	$ 15.55	3	20%	$5.44	Office Supplies	Paper	Xerox 1967
15	CA-2019-161389	IM-15070	98103	OFF-BI-10003656	$ 407.98	3	20%	$132.59	Office Supplies	Binders	Fellowes PB200 Plastic Comb Binding Machine
16	US-2018-118983	HP-14815	76106	OFF-AP-10002311	$ 68.81	5	80%	($123.86)	Office Supplies	Appliances	Holmes Replacement Filter for HEPA Air Cleaner, Very Large Room, HEPA Filter
17	US-2018-118983	HP-14815	76106	OFF-BI-10000756	$ 2.54	3	80%	($3.82)	Office Supplies	Binders	Storex DuraTech Recycled Plastic Frosted Binders
18	CA-2017-105893	PK-19075	53711	OFF-ST-10004186	$ 665.88	6	0%	$13.32	Office Supplies	Storage	Stur-D-Stor Shelving, Vertical 5-Shelf: 72"H x 36"W x 18 1/2"D
19	CA-2017-167164	AG-10270	84084	OFF-ST-10000107	$ 55.50	2	0%	$9.99	Office Supplies	Storage	Fellowes Super Stor/Drawer
20	CA-2017-143336	ZD-21925	94109	OFF-AR-10003056	$ 8.56	2	0%	$2.48	Office Supplies	Art	Newell 341
21	CA-2017-143336	ZD-21925	94109	TEC-PH-10001949	$ 213.48	3	20%	$16.01	Technology	Phones	Cisco SPA 501G IP Phone
22	CA-2017-143336	ZD-21925	94109	OFF-BI-10002215	$ 22.72	4	20%	$7.38	Office Supplies	Binders	Wilson Jones Hanging View Binder, White, 1"
23	CA-2019-137330	KB-16585	68025	OFF-AR-10000246	$ 19.46	7	0%	$5.06	Office Supplies	Art	Newell 318

Sheet1

Ubah format data pada kolom sales dan profit menjadi currency dengan mata uang dollar dan maksimal dua angka di belakang koma

Jawaban yang di kotak hijau



	A	B	C	D	E	F	G	H	I	J	K
1	order_id	customer_id	postal_code	product_id	sales	quantity	discount	profit	category	subcategory	product_name
2	CA-2019-152156	CG-12520	42420	FUR-BO-10001798	$ 261.96	2	0%	$41.91	Furniture	Bookcases	Bush Somerset Collection Bookcase
3	CA-2019-152156	CG-12520	42420	FUR-CH-10000454	$ 731.94	3	0%	$219.58	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs, Rounded Back
4	CA-2019-138688	DV-13045	90036	OFF-LA-10000240	$ 14.62	2	0%	$6.87	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters by Universal
5	US-2018-108966	SO-20335	33311	FUR-TA-10000577	$ 957.58	5	45%	($383.03)	Furniture	Tables	Bretford CR4500 Series Slim Rectangular Table
6	US-2018-108966	SO-20335	33311	OFF-ST-10000760	$ 22.37	2	20%	$2.52	Office Supplies	Storage	Eldon Fold 'N Roll Cart System
7	CA-2017-115812	BH-11710	90032	FUR-FU-10001487	$ 48.86	7	0%	$14.17	Furniture	Furnishings	Eldon Expressions Wood and Plastic Desk Accessories, Cherry Wood
8	CA-2017-115812	BH-11710	90032	OFF-AR-10002833	$ 7.28	4	0%	$1.97	Office Supplies	Art	Newell 322
9	CA-2017-115812	BH-11710	90032	TEC-PH-10002275	$ 907.15	6	20%	$90.72	Technology	Phones	Mitel 5320 IP Phone VoIP phone
10	CA-2017-115812	BH-11710	90032	OFF-BI-10003910	$ 18.50	3	20%	$5.78	Office Supplies	Binders	DXL Angle-View Binders with Locking Rings by Samsill
11	CA-2017-115812	BH-11710	90032	OFF-AP-10002892	$ 114.90	5	0%	$34.47	Office Supplies	Appliances	Belkin F5C206VTEL 6 Outlet Surge
12	CA-2017-115812	BH-11710	90032	FUR-TA-10001539	$ 1,706.18	9	20%	$85.31	Furniture	Tables	Chromcraft Rectangular Conference Tables
13	CA-2017-115812	BH-11710	90032	TEC-PH-10002033	$ 911.42	4	20%	$68.36	Technology	Phones	Konftel 250 Conference phone - Charcoal black
14	CA-2020-114412	AA-10480	28027	OFF-PA-10002365	$ 15.55	3	20%	$5.44	Office Supplies	Paper	Xerox 1967
15	CA-2019-161389	IM-15070	98103	OFF-BI-10003656	$ 407.98	3	20%	$132.59	Office Supplies	Binders	Fellowes PB200 Plastic Comb Binding Machine
16	US-2018-118983	HP-14815	76106	OFF-AP-10002311	$ 68.81	5	80%	($123.86)	Office Supplies	Appliances	Holmes Replacement Filter for HEPA Air Cleaner, Very Large Room, HEPA Filter
17	US-2018-118983	HP-14815	76106	OFF-BI-10000756	$ 2.54	3	80%	($3.82)	Office Supplies	Binders	Storex DuraTech Recycled Plastic Frosted Binders
18	CA-2017-105893	PK-19075	53711	OFF-ST-10004186	$ 665.88	6	0%	$13.32	Office Supplies	Storage	Stur-D-Stor Shelving, Vertical 5-Shelf: 72"H x 36"W x 18 1/2"D
19	CA-2017-167164	AG-10270	84084	OFF-ST-10000107	$ 55.50	2	0%	$9.99	Office Supplies	Storage	Fellowes Super Stor/Drawer
20	CA-2017-143336	ZD-21925	94109	OFF-AR-10003056	$ 8.56	2	0%	$2.48	Office Supplies	Art	Newell 341
21	CA-2017-143336	ZD-21925	94109	TEC-PH-10001949	$ 213.48	3	20%	$16.01	Technology	Phones	Cisco SPA 501G IP Phone
22	CA-2017-143336	ZD-21925	94109	OFF-BI-10002215	$ 22.72	4	20%	$7.38	Office Supplies	Binders	Wilson Jones Hanging View Binder, White, 1"
23	CA-2019-137330	KB-16585	68025	OFF-AR-10000246	$ 19.46	7	0%	$5.06	Office Supplies	Art	Newell 318

Pada kolom profit tandai profit yang negatif dengan warna merah

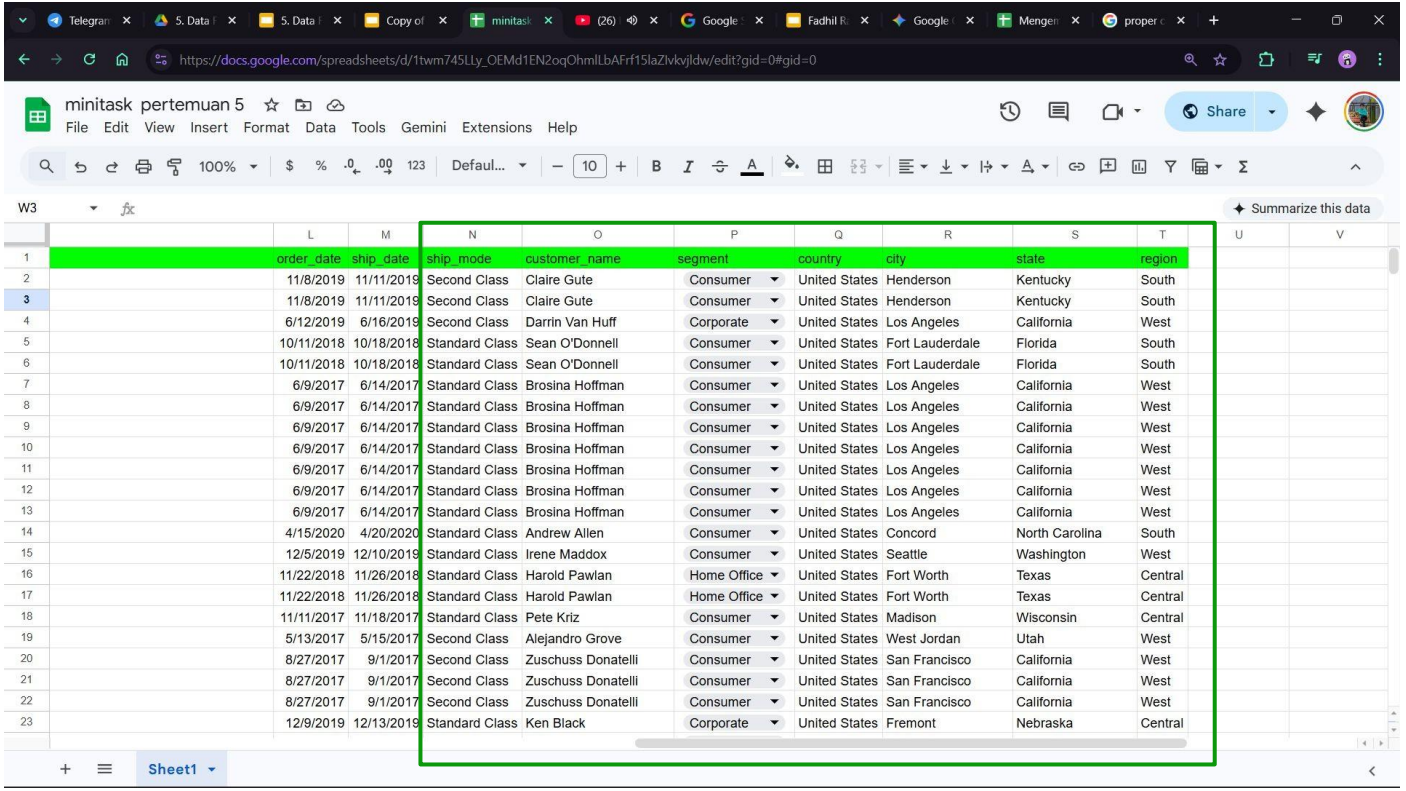
Jawaban yang di kotak hijau

minitask pertemuan 5

order_id	customer_id	postal_code	product_id	sales	quantity	discount	profit	category	subcategory	product_name
CA-2019-152156	CG-12520	42420	FUR-BO-10001798	$ 261.96	2	0%	$41.91	Furniture	Bookcases	Bush Somerset Collection Bookcase
CA-2019-152156	CG-12520	42420	FUR-CH-10000454	$ 731.94	3	0%	$219.58	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs, Rounded Back
CA-2019-138688	DV-13045	90036	OFF-LA-10000240	$ 14.62	2	0%	$6.87	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters by Universal
US-2018-108966	SO-20335	33311	FUR-TA-10000577	$ 957.58	5	45%	($383.03)	Furniture	Tables	Bretford CR4500 Series Slim Rectangular Table
US-2018-108966	SO-20335	33311	OFF-ST-10000760	$ 22.37	2	20%	$2.52	Office Supplies	Storage	Eldon Fold 'N Roll Cart System
CA-2017-115812	BH-11710	90032	FUR-FU-10001487	$ 48.86	7	0%	$14.17	Furniture	Furnishings	Eldon Expressions Wood and Plastic Desk Accessories, Cherry Wood
CA-2017-115812	BH-11710	90032	OFF-AR-10002833	$ 7.28	4	0%	$1.97	Office Supplies	Art	Newell 322
CA-2017-115812	BH-11710	90032	TEC-PH-10002275	$ 907.15	6	20%	$90.72	Technology	Phones	Mitel 5320 IP Phone VoIP phone
CA-2017-115812	BH-11710	90032	OFF-BI-10003910	$ 18.50	3	20%	$5.78	Office Supplies	Binders	DXL Angle-View Binders with Locking Rings by Samsill
CA-2017-115812	BH-11710	90032	OFF-AP-10002892	$ 114.90	5	0%	$34.47	Office Supplies	Appliances	Belkin F5C206VTEL 6 Outlet Surge
CA-2017-115812	BH-11710	90032	FUR-TA-10001539	$ 1,706.18	9	20%	$85.31	Furniture	Tables	Chromcraft Rectangular Conference Tables
CA-2017-115812	BH-11710	90032	TEC-PH-10002033	$ 911.42	4	20%	$68.36	Technology	Phones	Konftel 250 Conference phone - Charcoal black
CA-2020-114412	AA-10480	28027	OFF-PA-10002365	$ 15.55	3	20%	$5.44	Office Supplies	Paper	Xerox 1967
CA-2019-161389	IM-15070	98103	OFF-BI-10003656	$ 407.98	3	20%	$132.59	Office Supplies	Binders	Fellowes PB200 Plastic Comb Binding Machine
US-2018-118983	HP-14815	76106	OFF-AP-10002311	$ 68.81	5	80%	($123.86)	Office Supplies	Appliances	Holmes Replacement Filter for HEPA Air Cleaner, Very Large Room, HEPA Filter
US-2018-118983	HP-14815	76106	OFF-BI-10000756	$ 2.54	3	80%	($3.82)	Office Supplies	Binders	Storex DuraTech Recycled Plastic Frosted Binders
CA-2017-105893	PK-19075	53711	OFF-ST-10004186	$ 665.88	6	0%	$13.32	Office Supplies	Storage	Stur-D-Stor Shelving, Vertical 5-Shelf: 72"H x 36"W x 18 1/2"D
CA-2017-167164	AG-10270	84084	OFF-ST-10000107	$ 55.50	2	0%	$9.99	Office Supplies	Storage	Fellowes Super Stor/Drawer
CA-2017-143336	ZD-21925	94109	OFF-AR-10003056	$ 8.56	2	0%	$2.48	Office Supplies	Art	Newell 341
CA-2017-143336	ZD-21925	94109	TEC-PH-10001949	$ 213.48	3	20%	$16.01	Technology	Phones	Cisco SPA 501G IP Phone
CA-2017-143336	ZD-21925	94109	OFF-BI-10002215	$ 22.72	4	20%	$7.38	Office Supplies	Binders	Wilson Jones Hanging View Binder, White, 1"
CA-2019-137330	KB-16585	68025	OFF-AR-10000246	$ 19.46	7	0%	$5.06	Office Supplies	Art	Newell 318

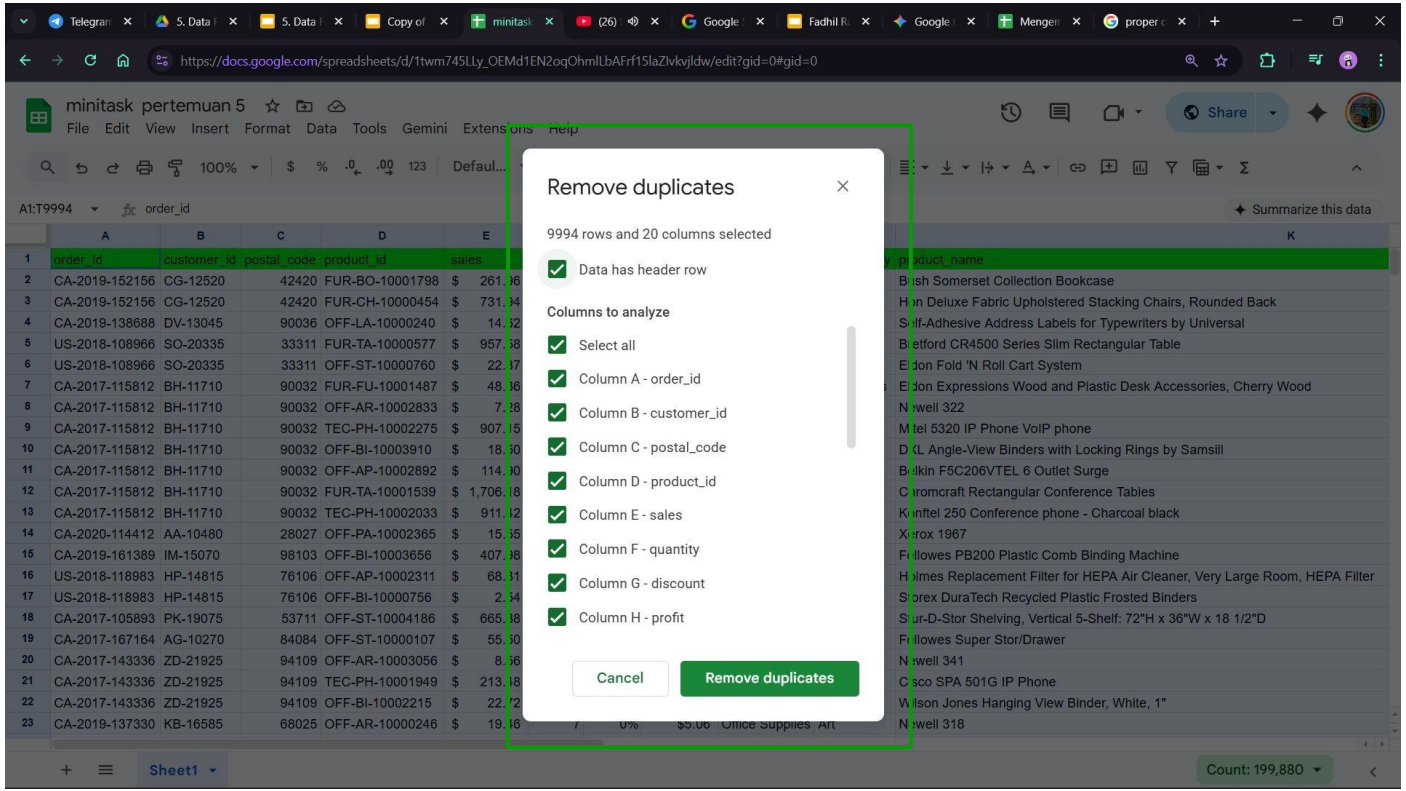
Ubah data pada kolom Ship_Mode sampai Region dengan Proper Case

Jawaban yang di kotak hijau



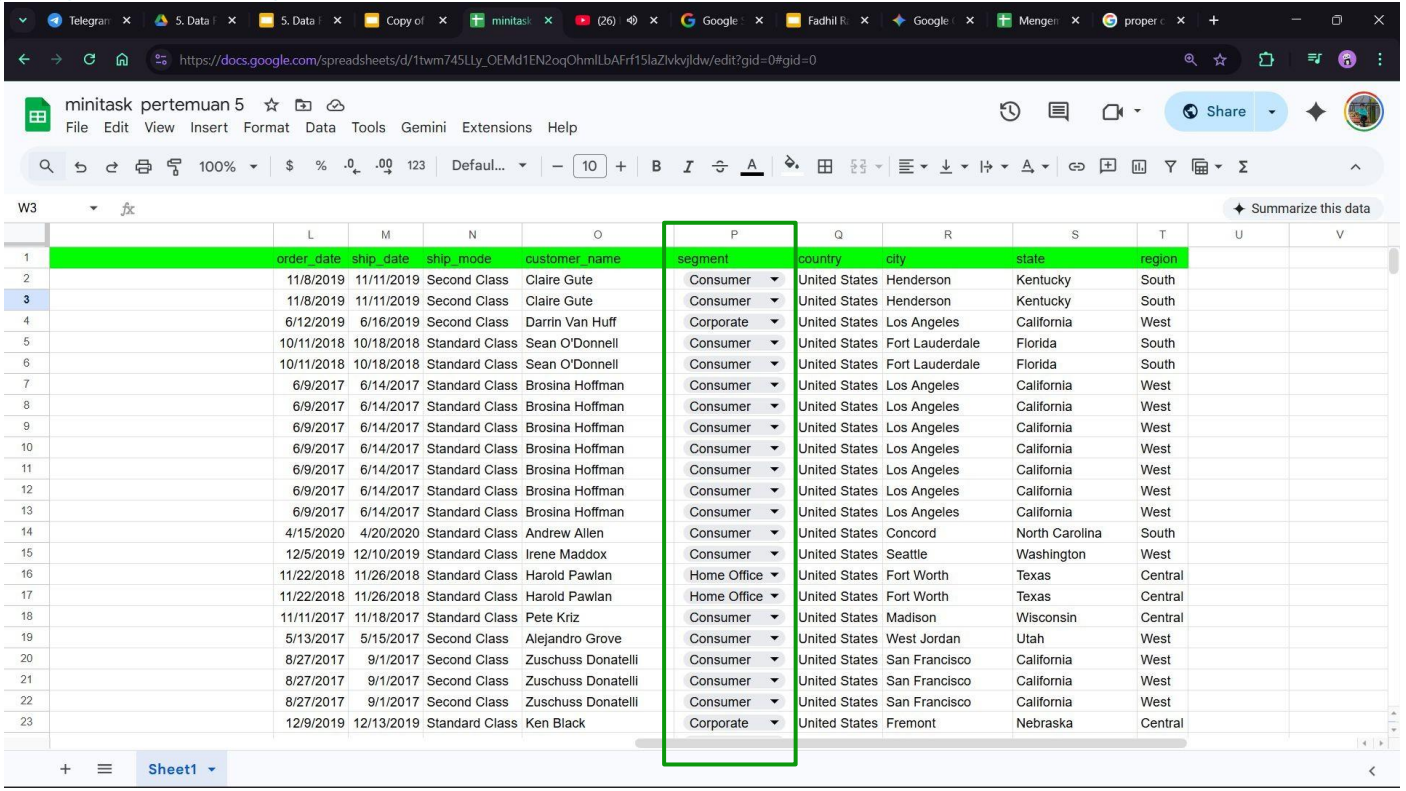
	order_date	ship_date	ship_mode	customer_name	segment	country	city	state	region
2	11/8/2019	11/11/2019	Second Class	Claire Gute	Consumer	United States	Henderson	Kentucky	South
3	11/8/2019	11/11/2019	Second Class	Claire Gute	Consumer	United States	Henderson	Kentucky	South
4	6/12/2019	6/16/2019	Second Class	Darrin Van Huff	Corporate	United States	Los Angeles	California	West
5	10/11/2018	10/18/2018	Standard Class	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	South
6	10/11/2018	10/18/2018	Standard Class	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	South
7	6/9/2017	6/14/2017	Standard Class	Brosina Hoffman	Consumer	United States	Los Angeles	California	West
8	6/9/2017	6/14/2017	Standard Class	Brosina Hoffman	Consumer	United States	Los Angeles	California	West
9	6/9/2017	6/14/2017	Standard Class	Brosina Hoffman	Consumer	United States	Los Angeles	California	West
10	6/9/2017	6/14/2017	Standard Class	Brosina Hoffman	Consumer	United States	Los Angeles	California	West
11	6/9/2017	6/14/2017	Standard Class	Brosina Hoffman	Consumer	United States	Los Angeles	California	West
12	6/9/2017	6/14/2017	Standard Class	Brosina Hoffman	Consumer	United States	Los Angeles	California	West
13	6/9/2017	6/14/2017	Standard Class	Brosina Hoffman	Consumer	United States	Los Angeles	California	West
14	4/15/2020	4/20/2020	Standard Class	Andrew Allen	Consumer	United States	Concord	North Carolina	South
15	12/5/2019	12/10/2019	Standard Class	Irene Maddox	Consumer	United States	Seattle	Washington	West
16	11/22/2018	11/26/2018	Standard Class	Harold Pawlan	Home Office	United States	Fort Worth	Texas	Central
17	11/22/2018	11/26/2018	Standard Class	Harold Pawlan	Home Office	United States	Fort Worth	Texas	Central
18	11/11/2017	11/18/2017	Standard Class	Pete Kriz	Consumer	United States	Madison	Wisconsin	Central
19	5/13/2017	5/15/2017	Second Class	Alejandro Grove	Consumer	United States	West Jordan	Utah	West
20	8/27/2017	9/1/2017	Second Class	Zuschuss Donatelli	Consumer	United States	San Francisco	California	West
21	8/27/2017	9/1/2017	Second Class	Zuschuss Donatelli	Consumer	United States	San Francisco	California	West
22	8/27/2017	9/1/2017	Second Class	Zuschuss Donatelli	Consumer	United States	San Francisco	California	West
23	12/9/2019	12/13/2019	Standard Class	Ken Black	Corporate	United States	Fremont	Nebraska	Central

Periksa apakah data superstore memiliki transaksi duplikat
Jawaban yang di kotak hijau



Pada bagian Segment terapkan data validation sehingga data yang masuk adalah satu dari pilihan berikut: Corporate, Consumer, Home Office

Jawaban yang di kotak hijau



	L	M	N	O	P	Q	R	S	T
1	order_date	ship_date	ship_mode	customer_name	segment	country	city	state	region
2	11/8/2019	11/11/2019	Second Class	Claire Gute	Consumer	United States	Henderson	Kentucky	South
3	11/8/2019	11/11/2019	Second Class	Claire Gute	Consumer	United States	Henderson	Kentucky	South
4	6/12/2019	6/16/2019	Second Class	Darrin Van Huff	Corporate	United States	Los Angeles	California	West
5	10/11/2018	10/18/2018	Standard Class	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	South
6	10/11/2018	10/18/2018	Standard Class	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	South
7	6/9/2017	6/14/2017	Standard Class	Brosina Hoffman	Consumer	United States	Los Angeles	California	West
8	6/9/2017	6/14/2017	Standard Class	Brosina Hoffman	Consumer	United States	Los Angeles	California	West
9	6/9/2017	6/14/2017	Standard Class	Brosina Hoffman	Consumer	United States	Los Angeles	California	West
10	6/9/2017	6/14/2017	Standard Class	Brosina Hoffman	Consumer	United States	Los Angeles	California	West
11	6/9/2017	6/14/2017	Standard Class	Brosina Hoffman	Consumer	United States	Los Angeles	California	West
12	6/9/2017	6/14/2017	Standard Class	Brosina Hoffman	Consumer	United States	Los Angeles	California	West
13	6/9/2017	6/14/2017	Standard Class	Brosina Hoffman	Consumer	United States	Los Angeles	California	West
14	4/15/2020	4/20/2020	Standard Class	Andrew Allen	Consumer	United States	Concord	North Carolina	South
15	12/5/2019	12/10/2019	Standard Class	Irene Maddox	Consumer	United States	Seattle	Washington	West
16	11/22/2018	11/26/2018	Standard Class	Harold Pawlan	Home Office	United States	Fort Worth	Texas	Central
17	11/22/2018	11/26/2018	Standard Class	Harold Pawlan	Home Office	United States	Fort Worth	Texas	Central
18	11/11/2017	11/18/2017	Standard Class	Pete Kriz	Consumer	United States	Madison	Wisconsin	Central
19	5/13/2017	5/15/2017	Second Class	Alejandro Grove	Consumer	United States	West Jordan	Utah	West
20	8/27/2017	9/1/2017	Second Class	Zuschuss Donatelli	Consumer	United States	San Francisco	California	West
21	8/27/2017	9/1/2017	Second Class	Zuschuss Donatelli	Consumer	United States	San Francisco	California	West
22	8/27/2017	9/1/2017	Second Class	Zuschuss Donatelli	Consumer	United States	San Francisco	California	West
23	12/9/2019	12/13/2019	Standard Class	Ken Black	Corporate	United States	Fremont	Nebraska	Central